

Rizal Kurnia

+62-851-9129-4341 | rizalkurnia.me@gmail.com | github.com/rizalkr | rizalkurnia.my.id

EDUCATION

Universitas Semarang

Semarang, Indonesia

Bachelor of Computer Science (S1 Teknik Informatika) | GPA: 3.77/4.00

Aug. 2022 – June 2026

- **Thesis (Grade: A):** "Komparasi Algoritma Content-Based Filtering dan Item-Based Collaborative Filtering pada Rekomendasi Indekos".
- Conducted an algorithmic benchmark comparing Content-Based Filtering and Item-Based Collaborative Filtering to evaluate recommendation accuracy and system performance.
- **Relevant Coursework:** Database Systems, Framework Programming, Data Mining, Machine Learning Fundamentals, Cloud Computing, Web Engineering.

EXPERIENCE

Backend Developer Intern

June 2025

Diskominfo Semarang

Semarang, Indonesia

- Engineered a backend system for a local primary school to manage operational data.
- Translated non-technical requirements into a structured relational database schema and deployed the final application.

Full-Stack Developer (Community Service Program)

Oct. 2025

Kelurahan Cepoko

Semarang, Indonesia

- Implemented a web-based decision support system to identify and prioritize local UMKM products.

PROJECTS

KOSERA | *Next.js, Python, FastAPI, Sentence Transformers, Microservices*

- Engineered a smart property recommendation platform utilizing a microservices architecture to isolate the frontend from heavy AI computations.
- Integrated Sentence Transformers to enable Semantic Search, allowing the system to translate natural language contexts into spatial vector calculations.

TitipYuk Semarang | *Next.js 14, TypeScript, Supabase, Tailwind CSS*

- Built a secure item storage platform with complete authentication, protected routes, and session management using Supabase.
- Engineered a custom OTP verification system utilizing bcrypt hashing and rate-limiting (max attempts) to mitigate brute-force vulnerabilities.
- Integrated an AI chatbot (Lunos) with real-time streaming capabilities and persistent conversation history.

KRZ DIGISHOP | *Astro.js, CSS Grid, UI/UX Design*

- Developed a highly optimized, zero-JS landing page for premium digital solutions, ensuring lightning-fast loading times and adherence to Core Web Vitals.

TECHNICAL SKILLS & CERTIFICATIONS

Languages: TypeScript, JavaScript, Python, SQL, HTML/CSS

Frameworks & Libraries: Next.js, React, Astro.js, Node.js, FastAPI, Tailwind CSS

Developer Tools: Git, Docker, Supabase, Vercel, Ubuntu/Linux

Certifications: IT Specialist - Databases (Dec 2025)